202200130077 薛润泽

1. B

4. (1) $P(\text{没有同号}) = \frac{13\times12\times11}{13\times13\times13} = \frac{132}{169}$

(2) $P(\text{有同号}) = 1 - P(\text{没有同号}) = \frac{37}{169}$

5. (1) $P(\text{只订A报的}) = P(A\bar{B}\bar{C})$

$= P(A\overline{BUC}) = P(A - A(BUC))$

$= P(A) - P(ABUAC)$

$= P(A) - P(AB) - P(AC) + P(ABC)$

$= 45\% - 10\% - 8\% + 3\%$

$= 30\%$

(2) $P(\text{只订A与B报的}) = P(AB\bar{C})$

$= P(AB - ABC)$

$= P(AB) - P(ABC)$

$= 10\% - 3\% = 7\%$.

(3) $P(\text{只订一种报的})$

$= P(A\bar{B}\bar{C}) + P(\bar{A}B\bar{C}) + P(\bar{A}\bar{B}C)$

$= P(A) - P(AB) - P(AC) + P(ABC)$

$+ P(B) - P(AB) - P(BC) + P(ABC)$

$+ P(C) - P(AC) - P(BC) + P(ABC)$

$= 73\%$

(4) $P(\text{恰好订两种报的})$

$= P(AB\bar{C}) + P(A\bar{B}C) + P(\bar{A}BC)$

$= P(AB - ABC) + P(AC - ABC) + P(BC - ABC)$

$= P(AB) + P(BC) + P(BC) - 3P(ABC)$

$= 10\% + 8\% + 5\% - 3\times3\%$

$= 14\%$.

(5) $P(\text{至少订一种报的})$

$= P(AUBUC) = P(A) + P(B) + P(C)$

$- P(AB) - P(AC) - P(BC) + P(ABC)$

$= 45\% + 35\% + 30\% - 10\% - 8\% - 5\% + 3\%$

$= 90\%$.

(6) $P(\text{一个不订任何报的}) = 1 - P(\text{至少订一种报的})$

$= 10\%$

6. (1) $P(\text{最小号码为5}) = \frac{C_5^2}{C_{10}^3} = \frac{1}{12}$

(2) $P(\text{最大号码为5}) = \frac{C_4^2}{C_{10}^3} = \frac{1}{20}$

(3) $P(\text{中间号码为5}) = \frac{C_5^1 C_4^1}{C_{10}^3} = \frac{1}{6}$

9. (1) $P(\text{AA}) = \frac{200}{200+600+50} = \frac{4}{17}$

(2) $P(\text{AA或aa}) = \frac{200+50}{200+600+50} = \frac{5}{17}$

11. $P(\text{第三次才取得次品}) = \frac{90}{100} \cdot \frac{90}{100} \cdot \frac{89}{99} \cdot \frac{10}{98} = \frac{89}{1078} \approx 0.0826$

13. $P(\text{全部合格}) = (1-0.5\%)(1-0.2\%)(1-0.1\%)(1-0.6\%) \geq 98.4\%$

14. 设共取 n 只.

$(1-0.03)^n \geq 95\% \quad 0.05$

$0.97^n \geq 0.05$

$n \geq \log_{0.97} 0.05.$

15. $P(\text{颜色相同}) = P(\text{全红}) + P(\text{全白}) + P(\text{全黑})$

$= \frac{7}{25} \times \frac{6}{25} + \frac{3}{25} \times \frac{10}{25} + \frac{15}{25} \times \frac{9}{25} = \frac{207}{625} \approx 0.3312$

16. 设 A=取到正品. B=甲厂生产.

$P(B|A) = \frac{P(AB)}{P(A)} = \frac{P(A|B)P(B)}{P(A)}$

#16. 设 A=取到正品

$P(A) = \frac{5}{10} \times \frac{9}{10} + \frac{3}{10} \times \frac{14}{15} + \frac{2}{10} \times \frac{19}{20}$

$= 0.92$

设 B=甲厂生产.

$P(B|A) = \frac{P(AB)}{P(A)} = \frac{P(A|B) \cdot P(B)}{P(A)}$

$= \frac{\frac{9}{10} \cdot \frac{5}{10}}{\frac{92}{100}} = \frac{45}{92} \approx 0.4891$

17. $P(\text{第二次取出的都是新球}) = \frac{1}{C_{12}^3} \cdot \frac{C_9^3}{C_{12}^3} + \frac{C_9^1 \cdot C_3^2}{C_{12}^3} \cdot \frac{C_8^3}{C_{12}^3} + \frac{C_9^2 C_3^1}{C_{12}^3} \cdot \frac{C_7^3}{C_{12}^3} + \frac{C_9^3}{C_{12}^3} \cdot \frac{C_6^3}{C_{12}^3} = \frac{441}{3025}$

设 A_i=第 i 次取出来的都是新球.

$P(A_1|A_2) = \frac{P(A_1A_2)}{P(A_2)} = \frac{P(A_2|A_1)P(A_1)}{P(A_2)}$

$= \frac{5}{21} \approx 0.238$

18. (1) P(他两人都命中) $= 0.8 \times 0.9 = 0.72$

(2) P(他们有一人命中) $= 0.8 \times (1-0.9) + (1-0.8) \times 0.9 = 0.26$

(3) P(他们人命中) $= 0.72 + 0.26 = 0.98$

(4) P(他两人都不中) $= 1 - 0.98 = 0.02$

19. 设 A=他能用到2000小时

B=他能用到3000小时

$P(B|A) = \frac{P(AB)}{P(A)} = \frac{0.85}{0.92} = \frac{85}{92} \approx 0.924$

20. (1) P(都没有坏) $= 0.8 \times 0.8 \times 0.8 = 0.512$

(2) P(坏一个) $= C_3^1 \cdot 0.2 \times 0.8 \times 0.8 = 0.384$

(3) P(最多只有一只损坏) $= 0.512 + 0.384 = 0.896$.

21. $P = C_{12}^9 (0.7)^9 (0.3)^3 + C_{12}^{10} (0.7)^{10} (0.3)^2 + C_{12}^{11} (0.7)^{11} \cdot 0.3 + C_{12}^{12} (0.7)^{12}$

$P = C_{12}^9 (0.7)^9 (0.3)^3 + C_{12}^{10} (0.7)^{10} (0.3)^2 + C_{12}^{11} (0.7)^{11} \cdot 0.3 + C_{12}^{12} (0.7)^{12}$

$= 0.2397 + 0.1678 + 0.0712 + 0.0138$

$= 0.4925$

22. (1) P(他恰有两人为O型) $= C_5^2 \cdot (0.46)^2 (1-0.46)^3 \approx 0.3332$

(2) P(他恰有三人为O型,两人为A型) $= C_5^3 \cdot (0.46)^3 (0.4)^2 \approx 0.1357$

(3) $P = (1-0.03)^3 = 0.8587$

24. P(他按对了) $= \frac{1}{5} + \frac{4}{5} \times \frac{1}{4} + \frac{4}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{3}{5}$

25. P(第取到白球)

$$= \frac{N}{N+M}\cdot\frac{n+1}{n+m+1} + \frac{M}{N+M}\cdot\frac{n}{n+m+1}$$

$$= \frac{Nn+Mn+N}{(N+M)(n+m+1)}$$

27. 设甲获胜

$$P(A) = P_1 + (1-P_1)(1-P_2)P(A)$$

得 $P(A) = \frac{P_1}{P_1+P_2-P_1P_2}$

B=乙获胜

$$P(B) = 1-P(A) = \frac{P_2-P_1P_2}{P_1+P_2-P_1P_2}$$

28. 设 A_i=两人都投进 i 个球.

$$P(A_0) = (1-0.7)^3\cdot(1-0.6)^3 = 0.001728$$

$$P(A_1) = C_3^1\cdot 0.7\cdot(1-0.7)^2\cdot C_3^1\cdot 0.6\cdot(1-0.6)^2$$

$$= 0.054432$$

$$P(A_2) = C_3^2\cdot 0.7^2\cdot(1-0.7)\cdot C_3^2(0.6)^2(1-0.6)$$

$$= 0.190512$$

$$P(A_3) = 0.7^3\cdot 0.6^3 = 0.074088.$$

$$P = P(A_0)+P(A_1)+P(A_2)+P(A_3)$$

$$= 0.32076$$